

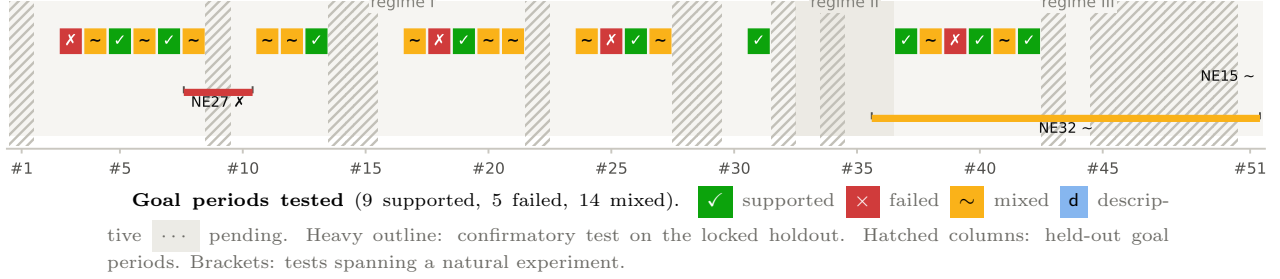
H82: Remanence: the endogenous part of the mean field

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 45% Faithfulness 1.5/5 Usefulness 1.5/5

Bottom line. The previous goal leaves a persistent trace on day-1 content ($\Delta\gamma_1 \approx 0.15$, 25 boundaries), but it is each agent's own carry-over, not a decaying remanence of the village record.

level: descriptive · pipeline checks 86% · updated 2026-10-04



WHAT HAPPENED

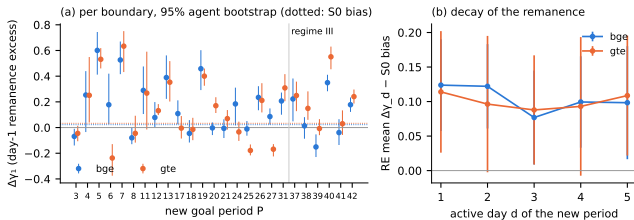
The test decomposes each agent's day-1 content in a new goal period into an exogenous field (new kickoff, goal text, previous kickoff, operator messages), the agent's prior from other periods, and the previous period's village centroid. It runs on 25 goal boundaries with both embedding models. The previous centroid loads by $\Delta\gamma_1 \approx 0.15$ beyond placebo centroids, 3 times the synthetic null. The trace does not decay over five days. The next period's centroid also loads, at about half the size. With the agent's own previous content as a regressor, the village centroid adds nothing ($-0.02 / -0.06$). Newcomers are too few to test. Read previous content predicts day-1 content better than unread content in one model. The holdout script exists and is not run.

PHYSICAL MODEL

Vector spins (model 11) with a two-part mean field. The spin $v_{i,d}$ is the agent-day content vector (32-d, style-residualized). X_{ex} are the exogenous directions. p_i is the agent's prior, fitted without periods P and $P-1$. $e_{P-1}^{(-i)}$ is the previous village centroid without agent i . Its loading γ is the endogenous field. The quench (H54) is the loading α on the new kickoff. Hysteresis is $\gamma > 0$ with $\gamma[P-1] > \gamma[P+1]$. A record-borne trace would also load in newcomers.

$$v_{i,d} = \sum_k \alpha_k X_{\text{ex},k} + \beta p_i^{(-P,-X)} + \gamma_d e_X^{(-i)} + \varepsilon_{i,d}$$

$$\Delta\gamma_d = \gamma_d[X=P-1] - \text{median}_{|Q-P|\geq 2} \gamma_d[X=Q], \quad A = \gamma_1[P-1] - \gamma_1[P+1]$$



Observables. (a) Day-1 remanence excess $\Delta\gamma_1$ per goal boundary with agent-bootstrap 95% intervals; dotted lines mark the synthetic null bias. Regime-I boundaries rest on 3–4 agents. (b) Random-effects mean of $\Delta\gamma_d$ minus the null bias over the first five active days. There is no decay.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

A goal change re-points agents onto the new kickoff, but each veteran keeps its own previous themes for at least five days. That carry-over is about 0.15 in loading units, beyond the new fields and the agent's prior. For an operator: a new goal does not erase the old one, so expect about a week of mixed content. Retire or reset a specific agent's context if its old thread must stop. Nothing here shows that the village record carries the trace on its own. The trace follows members and partly what they read.

RESULTS VS PREDICTIONS

Prediction	Observed (bge / gte)	Verdict
P1 $\Delta\gamma_1 > S_0$ q95	0.158 / 0.153 (q95 0.055 / 0.063); 17/25, 16/25 positive	pass
P2 asymmetry $A > S_0$ q95	0.093 / 0.104 (q95 0.036 / 0.052); part is lag geometry	pass
P3 decay within days	$\Delta\gamma_d$ 0.15, 0.15, 0.11, 0.13, 0.13	fail
P4 newcomers carry it	5 boundaries, 1 newcomer each; power ≤ 0.18	inconcl.
N1 NE27 empty-memory join	no trace of #8 for veterans or newcomers	fail
N2 NE32 isolated newcomers	+0.036 [0.006, 0.081] / +0.010 [-0.020, 0.039]	mixed
N3 NE15 read vs unread	+0.071 [-0.007, 0.148] / +0.106 [0.039, 0.174]	mixed
PH after kickoff only	0.152 / 0.156	post hoc
PH own past as regressor	-0.024 / -0.060	post hoc

GOAL PERIODS IN DETAIL

G03 $\times$ failed
G04 $\sim$ mixed
G05 $\checkmark$ supported
G06 $\sim$ mixed
G07 $\checkmark$ supported
G08 $\sim$ mixed
G11 $\sim$ mixed
G12 $\sim$ mixed
G13 $\checkmark$ supported
G17 $\sim$ mixed

+18 more periods: see the card.

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	2
B assumptions	0	G ground truth	1
C adequacy	2	H comparative	1
D unfitted predictions	1	I transfer	1
E interventional	1		

Level: **descriptive**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

Each boundary is one regression on 3–15 agents. The placebo-corrected excess is biased upward under the null (+0.02 / +0.03), so decisions use the synthetic percentile. The newcomer clause, the HH’s central claim, is untestable at this sample size. The own-past regressor also contains the village state as the agent saw it, so the PH2 result says which regressor wins, not which mechanism acts. Part of the asymmetry is the lag geometry of H81’s slow mode.

ROUND 2

Model each agent’s carry-over with a decay over the whole period and compare it with H81’s τ_u . Pool all newcomer first weeks against same-day veterans, after a synthetic power check. Extend the read-vs-unread test to memory and history-search reads.

- **Essence.** whether a goal change resets the village or only re-points members who keep their own threads.
- **H82-R1** Individual carry-over model.
- **H82-R2** The record channel.
- **H82-R3** Newcomer power.

Second figure. (a) Mean day-1 loading on the previous, a placebo and the next period’s centroid (19 boundaries with an eligible next period). (b) Synthetic power on the real boundaries: the pooled test reaches 0.83–0.92 at $\gamma = 0.05$; the newcomer test stays below 0.2.

